GBENGA S. DAIRO

PhD Student, Quantitative Biomedicine | Computational & Experimental Biology
Rutgers University, Piscataway, NJ | gsd67@scarletmail.rutgers.edu | Website | LinkedIn | GitHub | Google Scholar | ORCID

PROFESSIONAL SUMMARY

PhD student in Quantitative Biomedicine with interdisciplinary expertise in computational biology, experimental molecular biology, and translational research. Integrates genomics, epigenomics, metabolomics, and clinical phenotypes to investigate disease mechanisms and treatment response. Experienced in molecular QTL analysis, Bayesian colocalization, computational drug discovery, and experimental approaches involving in vivo disease models and molecular and cell-based assays. Develops reproducible R/Python workflows for multi-omics integration, statistical analysis, and biological data interpretation.

CORE SKILLS

Computational Biology & Multi-Omics: R | Python | Bash | Genomics | Epigenomics | Transcriptomics | RNA-seq Differential Expression | Molecular QTL Analysis | Bayesian Colocalization
Experimental Biology: Cell Culture | Cell-Based Assays | In Vivo Mouse Models | Cryostat Tissue Sectioning | PCR/Genotyping | Western Blotting | Immunostaining | Protein Extraction & Purification
Computational Drug Discovery: Molecular Docking | Molecular Dynamics | DFT | Schrödinger Suite | GROMACS | Gaussian 16
Quantitative & Data Science: Statistical Modeling | AI & Machine Learning | Longitudinal/Mixed-Effects Modeling | Data Visualization | Reproducible Analysis | Git/GitHub

EDUCATION

Ph.D., Quantitative Biomedicine, GPA: 3.79/4.00 2024 – Present
Institute of Quantitative Biomedicine, Rutgers University, New Brunswick, NJ

Graduate Certificate, Computational Biology (in progress) Present
Rutgers University, New Brunswick, NJ

M.S., Cell and Molecular Biology, GPA: 3.86/4.00 2023
Western Illinois University, Macomb, IL

B.S., Forestry and Wildlife Management, GPA: 3.52/4.00 2017
Federal University of Agriculture, Abeokuta, Nigeria

RESEARCH EXPERIENCE

Graduate Researcher Sep 2024 – Present
Kachroo Lab, Institute of Quantitative Biomedicine, Rutgers University

- Lead computational analyses integrating genomics, epigenomics, transcriptomics, metabolomics, and clinical phenotypes across pediatric disease cohorts to investigate molecular mechanisms and treatment response using R/Python-based bioinformatics and statistical workflows.
- Analyzed a targeted panel of 16 endogenous steroid metabolites in 664 children with asthma at baseline and 4-year follow-up using longitudinal statistical models to evaluate treatment, time, sex, and developmental effects; work published in Respiratory Research (2026).
- Analyze treatment-response relationships across clinical phenotypes and molecular endpoints, translating statistical findings into biological interpretation and publication-ready visualizations for multidisciplinary collaborators.
- Integrated genome-wide molecular QTL mapping and Bayesian colocalization across two independent cohorts to identify shared genetic signals underlying DNA methylation and metabolite associations, and investigated their relationships with asthma-related clinical phenotypes; research accepted for poster presentation at the 2026 American Society of Human Genetics Annual Meeting.
- Develop reproducible R/Python bioinformatics workflows and R Shiny applications for omics, pathway-enrichment, and interactive data exploration; maintain version-controlled code on GitHub.

Predoctoral Research Trainee Jun 2023 – Aug 2024
Hla Lab, Boston Children's Hospital / Harvard Medical School, Boston, MA

- Investigated sphingosine-1-phosphate signaling and sphingosine kinase 1 (SPHK1) regulation in ischemia-reperfusion injury using in vivo mouse models, including vena cava blood collection and downstream molecular analyses; contributed to a manuscript under review.
- Conducted sphingosine dose-response experiments in SPHK1 wild-type and knockout mouse red blood cells to investigate SPHK1-dependent hemolysis, observing hemolytic responses in wild-type but not knockout cells.
- Generated and analyzed experimental data using cryostat tissue sectioning, Western blotting, immunostaining, PCR/genotyping, gel electrophoresis, protein extraction and purification, and cell-based assays.

Graduate Research Assistant Jul 2022 – Apr 2023
Soendergaard & Determan Labs, Western Illinois University

- Applied molecular docking, molecular dynamics simulations, and density functional theory to investigate bioactive compounds targeting proteins implicated in ovarian and pancreatic cancers, integrating structural modeling with computational drug-discovery hypotheses; research contributed to peer-reviewed publications.
- Evaluated the cytotoxic effect of morel mushroom extract on MIA-PaCa 2 pancreatic cancer cell lines using MTT assays.

SELECTED PUBLICATIONS

Dairo, G., Sharma, R... & Kachroo, P. Metabolomic signatures of the steroid biosynthesis driving sex differences in clinical asthma subtypes. *Respir Res* (2026). https://doi.org/10.1186/s12931-026-03874-1.

Dairo, G., Ward, M., Soendergaard, M, and Determan, J. Bioactive compounds from Morchella esculenta as potential inhibitors of RNA-binding protein La in ovarian cancer: a molecular modeling and quantum mechanics approach. *In Silico Pharmacology (2024).* https://doi.org/10.1007/s40203-024-00202-7.

Dairo, G., Ilesanmi, A., Balogun, T., Ward, M., Soendergaard, M., & Determan, J. Computational evaluation of bioactive compounds from *Viscum album* (mistletoe) as inhibitors of p63 for pancreatic cancer treatment. *Journal of Biomolecular Structure and Dynamics* (2023). https://doi.org/10.1080/07391102.2023.2191133.

COMMUNICATION & LEADERSHIP

- Presented research through oral and poster presentations at Rutgers scientific symposia and regional scientific meetings (2023–2026).
- Teaching Assistant, Western Illinois University (2021–2022): taught molecular biology and ecology, including hypothesis development, R-based statistical analysis, and laboratory methods.